University of Macau 2026–2027 Semester 1

Practice Final Examination

CIVL 7208/7014

Special Topics in CPM / Numerical Modeling in Geomechanics

Practice instructions

This practice paper contains more application and proof problems than the actual final examination, and they are generally more difficult. Work through the extensions as well: understanding the ideas behind these problems will be very helpful for the final. The problems are arranged roughly in order of increasing difficulty.

Multiple-choice and true/false questions are not included here. For multiple-choice questions, review the *Matrix Crime* slides. For true/false questions, review linear independence of vectors (slide 03) and matrix inverses (slide 08).

Use the Wenjuanxing self-test to check your understanding. Click the link below or scan the QR code. You may submit your responses. **After submitting, you will be able to view the correct answers.**
https://ks.wjx.com/vm/mPil1xS.aspx

Actual final examination: information and instructions

Date: November 24, 2026
Time: 19:00 to 22:00
Location: E4-1051

This is a 180-minute in-class final examination.

Hard copies of any related materials may be used during the exam. Electronic versions on computers or mobile phones are strictly prohibited. Please answer all questions. Some are (quite) straightforward. Others, not so much.

For multiple-choice and true/false questions, write your final answer **clearly and definitively**. No reasoning is required. An unclear or ambiguous answer will receive **zero marks** for that item.

For the remaining problems, show your work and keep your paper neat. Avoid making your solutions too messy.

Some problems involve applications, but you do not need prior knowledge of the application area to solve them. The problem statement contains everything you need.

The problems *usually* appear in order of increasing difficulty.

P1. *Fitting a piecewise affine function to data*

We are given samples (x_i, y_i), $i = 1, \ldots, N$, where x_i and y_i are scalars. We fit a function of the form

$$\widehat{y} = \widehat{f}(x) = \theta_1 + \theta_2 \min\{x, 0\} + \theta_3 \max\{x, 0\},$$

where $\theta = (\theta_1, \theta_2, \theta_3)^T$ contains the unknown parameters.

The function is affine for $x \leq 0$ and for $x \geq 0$, and it is continuous at $x = 0$ with value θ_1. Such a function is called *piecewise affine*, with a single knot or kink at zero. To see its shape, you may sketch it using, for example, $\theta = (1, 2, 3)^T$.

Consider the following five data points:

$$(-2, 3), \quad (-1, 1), \quad (0, 1), \quad (1, 2), \quad (1, 1).$$

Give the matrix A and vector b for which the sum of squared fitting errors equals $\|A\theta - b\|_2^2$. Write down the **explicit numerical entries** of A and b and specify their dimensions. Treat all five pairs as separate observations.

You do not need to solve the least-squares problem or calculate $\widehat{\theta}$.

Answer:

[Continue your answers here.]

P2. *Estimating a ball's initial velocity*

Consider a ball moving in two dimensions, ignoring air resistance. Its initial position is known to be $p_0 = \mathbf{0}$, and its unknown initial velocity is $v \in \mathbb{R}^2$, with horizontal and vertical components. Its position at time t is

$$p(t) = tv - \frac{a}{2}\begin{bmatrix}0\\t^2\end{bmatrix},$$

where $a > 0$ is the known gravitational acceleration. At known positive times $t_1, \ldots, t_k$ after launch, we measure noisy positions $p_1, \ldots, p_k \in \mathbb{R}^2$.

We estimate the initial velocity by choosing $\widehat{v}$ to minimize

$$\sum_{i=1}^{k} \left\| t_i v - \frac{a}{2}\begin{bmatrix}0\\t_i^2\end{bmatrix} - p_i \right\|_2^2$$

over v. Formulate this as a least-squares problem of minimizing $\|Av - b\|_2^2$. Give the matrix A and vector b and specify their dimensions. What is the minimum number of measurements k needed for a unique solution?

Hint. For vectors x, y, z, w of appropriate sizes,

$$\|x - y\|_2^2 + \|z - w\|_2^2 = \left\| \begin{bmatrix}x\\z\end{bmatrix} - \begin{bmatrix}y\\w\end{bmatrix} \right\|_2^2.$$

Extension: unknown initial position. Suppose the initial position $p_0 \in \mathbb{R}^2$ is also unknown, so that

$$p(t) = p_0 + tv - \frac{a}{2}\begin{bmatrix}0\\t^2\end{bmatrix}.$$

Can you still use least squares to estimate p_0 and v simultaneously? Formulate the problem, give the new matrix and vector and their dimensions, and determine the minimum number of measurements and the condition on the measurement times needed for a unique solution.

Answer:

[Continue your answers here.]

P3. *Finding physical laws with regression*

Many physical laws can be written as polynomial equations. For example, Newton's second law is $F = MA$ (force is mass times acceleration), the ideal gas law is $PV = NRT$ (pressure times volume equals the number of moles times the ideal gas constant times temperature), and kinetic energy is $K = \frac{1}{2}MV^2$, where M is mass and V is velocity.

Suppose we have k physical quantities represented by positive values $A_1, \ldots, A_k$, and we believe they are related by

$$1 = c\, A_1^{\beta_1} A_2^{\beta_2} \cdots A_k^{\beta_k},$$

where $c > 0$ and $\beta_1, \ldots, \beta_k \in \mathbb{R}$ are unknown scalars, with the exponents believed to be nonzero.

We are given m experimental measurements

$$a_i = (A_{1,i}, \ldots, A_{k,i}) \in \mathbb{R}^k_{++}, \qquad i = 1, \ldots, m,$$

so the jth component of a_i is $A_{j,i} > 0$.

How might you estimate a nontrivial set of parameters $c, \beta_1, \ldots, \beta_k$ from $a_1, \ldots, a_m$ by solving a least-squares problem? Explain any transformations of the problem data and give the dimensions of your least-squares problem. State and justify any additional choices you make to exclude the trivial solution. Different valid least-squares formulations are acceptable if these choices are clearly explained.

Hint. Think about transformations that linearize the equation $1 = c\, A_1^{\beta_1} \cdots A_k^{\beta_k}$. How can you avoid the trivial solution $c = 1$ and $\beta_1 = \cdots = \beta_k = 0$? Note that, for any nonzero scalar β, raising both sides of the equation to the power $1/\beta$ preserves the equality:

$$1 = \left(c\, A_1^{\beta_1} \cdots A_k^{\beta_k}\right)^{1/\beta} = c^{1/\beta} A_1^{\beta_1/\beta} \cdots A_k^{\beta_k/\beta}.$$

Answer:

[Continue your answers here.]

P4. *Block matrix inversion*

Let $A \in \mathbb{R}^{n\times n}$, $U \in \mathbb{R}^{n\times k}$, and $V \in \mathbb{R}^{k\times n}$, where A is invertible. Assume that $I_k + VA^{-1}U$ is invertible. Prove the Sherman–Morrison–Woodbury formula

$$(A+UV)^{-1} = A^{-1} - A^{-1}U(I_k + VA^{-1}U)^{-1}VA^{-1}. \tag{1}$$

Here, I_j denotes the $j \times j$ identity matrix.

Hint. Consider the block matrix equation

$$\begin{bmatrix} A & U \\ V & -I_k \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} I_n \\ 0_{k\times n} \end{bmatrix}, \qquad X \in \mathbb{R}^{n\times n}, \quad Y \in \mathbb{R}^{k\times n}.$$

Write this as two equations. Use the first equation to express X in terms of Y. Substitute into the second equation and solve for Y, then substitute back to obtain X.

Finally, use the second equation to express Y in terms of X, and substitute this expression into the first equation. What does the resulting equation tell you about X? Why does it imply that $X = (A+UV)^{-1}$?

Extension 1: a rank-one update. Replace U by a column vector $u \in \mathbb{R}^n$ and V by the transpose v^T of another column vector $v \in \mathbb{R}^n$. What formula do you obtain from (1)? State the condition under which this formula is valid.

Extension 2: direct verification. Verify the Sherman–Morrison–Woodbury formula (1) directly using the definition of a square matrix inverse. Let R denote the right-hand side of (1). Check both $(A+UV)R = I_n$ and $R(A+UV) = I_n$.

Reference for the rank-one formula: Sherman–Morrison formula.

Answer:

[Continue your answers here.]

P5. *Updating a least-squares solution with one additional row*

Challenging problem. This problem takes more work, but understanding the proof will be very helpful for the final examination.

Let $A \in \mathbb{R}^{m \times n}$ have linearly independent columns, and let $b \in \mathbb{R}^m$. The least-squares solution of minimizing $\|Ax - b\|_2^2$ is

$$\widehat{x} = (A^T A)^{-1} A^T b.$$

For a new observation given by a column vector $a \in \mathbb{R}^n$ and a scalar $b_{m+1} \in \mathbb{R}$, define

$$C = \begin{bmatrix} A \\ a^T \end{bmatrix} \in \mathbb{R}^{(m+1) \times n}, \qquad d = \begin{bmatrix} b \\ b_{m+1} \end{bmatrix} \in \mathbb{R}^{m+1}.$$

The updated least-squares solution is $\widehat{y} = (C^T C)^{-1} C^T d$. Prove the following rank-one update formula:

$$\widehat{y} = \widehat{x} + \frac{(A^T A)^{-1} a}{1 + a^T (A^T A)^{-1} a} \left(b_{m+1} - a^T \widehat{x} \right).$$

Suppose both $\widehat{x}$ and $(A^T A)^{-1}$ have already been computed. Roughly how many floating-point operations (flops), as a function of n, are required to compute $\widehat{y}$ using this update? Give the leading-order cost for a dense matrix.

Hint. Use the rank-one special case of the Sherman–Morrison–Woodbury formula from Problem P4.

Answer:

[Continue your answers here.]